Running pi with Docker Compose

https://gemini.google.com/app/5978ad61c5db5cc2

User prompt: can i run pi coding agent with docker compose?

Response: **Yes, you can run the Pi coding agent using Docker Compose.**

Many developers containerize Pi to establish a secure sandbox (often called "YOLO-mode" isolation) so that the agent can read, write, and execute code within your project directory without risking the rest of your host system.

What a Docker Compose Setup Looks Like

Typically, a Docker Compose configuration for Pi involves:

1. **Binds/Volume Mounts:** Mounting your local project directory into the container (e.g., at `/project` or `/home/node/app`) so Pi can manipulate the codebase.
2. **Persistent Config:** Mounting your host's `~/.pi` configuration and session directory so your API keys, settings, and history persist across container restarts.
3. **Environment Variables:** Passing through your LLM provider API keys (like `ANTHROPIC_API_KEY`, `OPENAI_API_KEY`, etc.).

Example Configuration

You can structure a basic `docker-compose.yml` file like this:

```
version: '3.8'

services:
  pi:
    build: .
    container_name: pi-coding-agent
    stdin_open: true
    tty: true
    environment:
      - ANTHROPIC_API_KEY=${ANTHROPIC_API_KEY}
    volumes:
      - .:/project
      - ~/.pi:/home/node/.pi
    working_dir: /project
```

And a companion `Dockerfile` to install the package:

```
FROM node:22-bookworm-slim

RUN apt-get update && apt-get install -y --no-install-recommends \
    git \
    ripgrep \
    curl \
    && rm -rf /var/lib/apt/lists/*
```

```
RUN npm install -g @earendil-works/pi-coding-agent

USER node
ENTRYPOINT ["pi"]
```

Running It

Once configured, you can build and start your interactive session using:

```
docker compose run --rm pi
```

Existing Community Projects

If you prefer not to write the configuration from scratch, several open-source templates and wrappers already set this up with advanced security features (like non-root user execution, network proxies, or token vaults):

- **`pi-less-yolo` / `pi-coding-agent-container` repositories on GitHub:** Offer production-ready Makefiles and Docker Compose setups tailored specifically for secure local Pi isolation.
- **`pi-in-a-box`:** A lightweight helper script ecosystem designed to spin up ephemeral or persistent Docker containers matching your current working directory.

Sources

1. pi-coding-agent docker compose - GitHub
2. GitHub - souldzin/pi-in-a-box: Dockerfile and script for running pi
3. Slightly less YOLO version of pi-coding-agent in Docker container

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